

Unit 6**Controlling****Structure:**

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6.1 Introduction

In the previous unit we dealt with the definition, comparison, and contrasting of directing and leading. We also discussed the characteristics, importance, and functions of leading.

Control is the last in the management process and is perhaps the most important. In reality, most managers are busy laying down controls and exercising them. Control is the rudder of business because it ensures that a process is going in the right direction by making continuous corrections. Without control, the business will perhaps go to where it should never go and do what it should never do. A business will be able to expand in product portfolio and geography and delegate and outsource only because controls are possible and in fact real time control can be exercised through IT. You may wonder how large systems like the East India Company expanded without IT in the earlier days. Even there, information was the hub of control. Perhaps, they did it using telegraph and later telephone, but today, it is done by computers. In this unit, we will deal with control and understand how it is exercised, and how IT is helping in controlling in a big way.

Objectives:

- To explain control and describe its importance
- To identify critical control points
- To analyse control as a feedback system
- To describe prerequisites of effective control
- To elucidate various control techniques
- To explain IT enabled 'controls and its challenges

Learning Outcomes:

After studying this unit, you will be able to:

- Define control and describe its importance.
- Explain the controlling process.
- Identify critical control points.
- State how control is used as a feedback system.
- Enumerate Prerequisites of Effective Control
- Explain various Control Techniques
- Explain IT Enabled 'Controls' and its Challenges

6.2 Case-let**Sambhavi Bakers Part III**

As the consultant left her office, Mrs. Rakhi busied herself by preparing the birthday cake for her daughter, Reshma, which she always baked herself. Reshma had grown, completed her MBA programme, worked in her production and outlets. She did not have much hesitation to dirty her hands. As they sat down the next day to cut the cake, their conversation moved to when she should get married. Reshma quickly raised the issue of her wanting to study more of baking and how she had quietly prepared and cleared her examinations for another masters in Cornell, which is known for its hospitality management. Reluctantly, Mrs. Rakhi took the decision, hard one as she would be away from her daughter for the first time. Nevertheless, the need for her daughter's growth overcame her reluctance and she let Reshma go.

Having graduated and worked in some of the state of art highly differentiated baking companies in the U.S and having worked part time in the bakery chains and retain chains, Reshma returned after six years,

confident and ready to have a go on her own. Her mother declined and told her to take over the business immediately and that she would withdraw completely in two years.

In the next phase of expansion, Reshma planned to make Sambhavi a national brand in five years and planned a major layout. She knew the heart of business i.e., quality and she knew from her education that standards are needed for everything. Standards for capital investment, per employee return, per customer revenue, and in fact standards that cover the production and every aspect of business. Like most young people she was computer savvy and wanted to use IT in a big way for controls and for entering both E-commerce and M-commerce. How wonderful it would be when you order a special cake for your mother's wedding anniversary even as you are driving home from a busy day. She knew the wind that was lashing the corporate India everywhere- in metros and small cities alike. She coined a nice brand name 'love bake' to cater for her special birthday cakes that she dreamt would be cut in most modern families in a decade from now. The target was hundred outlets per large state with a dozen in each metro city i.e., 150 outlets a year. She mused as she sat near her mother and signed the papers taking charge of Sambhavi Bakers.

Reshma then turned and called her consultant, this time not to discuss about the expansion, but to lay out the control plan of her business.

6.3 Definition, Importance, and Process of Controlling

Controlling can be defined as measuring and correcting of performance to achieve the organisational goals. According to Brech, "Controlling is a systematic exercise which is called as a process of checking actual performance against the standards or plans with a view to ensure adequate progress and also recording such experience as is gained as a contribution to possible future needs." According to Donnell, "Just as a navigator continually takes reading to ensure whether he is relative to a planned action, so should a business manager continually take reading to assure himself that his enterprise is on right course."

It is often considered as a part of planning. Though planning and controlling can be considered as two sides of a coin, it is possible for us to make a clear distinction between them because control has to be done against some parameter, which is laid down by planning. The following is the inter relation between them.

- Planning precedes controlling and controlling succeeds planning.
- Planning and controlling are inseparable functions of management.
- Activities are put on rails by planning and they are kept at a right place through controlling.
- The process of planning and controlling works on systems approach, which is as follows:
Planning → Results → Corrective Action (through control)
- Planning and controlling are integral parts of an organisation as both are important for smooth running of an enterprise. Planning and controlling reinforce each other. Each drives the other function of management.

Importance of control

Control is important because of the following reasons:

- ***Creates the basis for the future*** – It provides a basis for future planning and action because it identifies and reports the efficiency of a project or any action.
- ***Guides to keep goals on track*** – The continuous flow of information about projects and performance keeps the goals on track by enabling timely corrective action. Thus, it guides the management in achieving predetermined goals.
- ***Prevents repetition of mistakes*** – It also enables the management to avoid repetition of past mistakes by enabling one to decide on the future course of action.
- ***Enables future planning and follow up action*** – It enables planning for the future and facilitates follow up action.
- ***Enables decentralisation*** – In large organisations where decentralisation is inevitable, control becomes the key to success because it establishes accountability. The management cannot delegate authority without ensuring proper control. The targets or goals of various departments are used as a control technique.

- **Enable management by exception** – If the work is satisfactory then the top management should not feel worried. The management by exception enables top management to concentrate on policy formulation. Various control techniques like budgeting, cost control, pre-action approvals, etc. allow decentralisation without losing control over activities.
- **Assists co-ordination** – Control helps in co-ordination of activities through unity of action. Every manager will try to co-ordinate the activities of his/her subordinates in order to achieve departmental goals. Similarly, the chief executives also co-ordinate the functioning of various departments. The control acts as a check on the performance and proper results are achieved only when activities are co-ordinated.
- **Enables HR functions and motivation** – Various control devices act as motivators to managers as they measure their performance, which itself is a motivator. Besides, it ensures equity in reward, training, performance evaluation, promotion, and a plethora of other HR functions. It automatically leads to improvement of performance, which enables career progression and also prevents arbitrary decisions of many HR functions.

Process (steps) of controlling

1. **Establishment of standards** – Standards can be defined as selected points that are critical in execution of a plan measurement, which identifies the efficiency of execution of the plan. Standards are criterion for performance and can be divided into the following two types:
 - **Tangible standards** – These standards can be measured, such as profit, return on investment, rejection rate, customer complaints, etc.
 - **Intangibles standards** – These include attitude of workers, job satisfaction, managerial effectiveness, etc. Some caution is required when we call them intangible as explained below.
2. **Measurement of performance** – ‘What is measured cannot be improved’ is a well known axiom. Measurement is necessary for finding out deviations and is perhaps the most difficult step in controlling. Measurement of tangible standards is easy as it can be expressed in units, cost, money terms, etc. Quantitative measurement becomes difficult when performance of manager. Though literature tends to suggest that items such as attitude of the workers, their morale towards work, the development in the attitudes regarding the physical

environment, communication efficacy, etc. cannot be measured, this is a misnomer. Attitude surveys, job satisfaction scales, morale scales, etc. do exist and are used by several large companies. These measures are different from measuring in terms of specific financial performance, or quantitative output. In many ways, the focus on intangible measures of performance is a hangover of the past.

3. **Comparison of actual and standard performance** – Deviation can be defined as the gap between actual performance and the planned targets. Hence, comparison of actual performance with the planned targets is very important. We have to find out the extent of deviation and the cause of deviation. Extent of deviation means whether the deviation is positive or negative, whether the actual performance is in conformity with the planned performance and the degree of variance, if any. Once the deviation is identified, a manager has to think about various causes, which have led to deviations such as erroneous planning, lack of co-ordination, implementation defect, supervision and communication ineffectiveness, etc.
4. **Taking remedial actions** – Once the causes and extent of deviations are known, the manager has to detect those errors and take remedial measures. First is by taking corrective measures for deviations which have occurred; and second is by evaluating the target or standard itself because if things are not rectified even after corrective actions, it could be due to erroneous target. However, one should take the second option only with due caution.

Self Assessment Questions

1. Controlling can be defined as _____ and _____ of performance to achieve the organisational goals.
2. Among various reasons why controlling is important, guiding to keep _____ on track is a significant one.
3. There are four steps in the process of controlling. These are measurement of performance, comparison of actual and standard performance, taking remedial actions, and creating tangible measures. (True/False)
4. 'Rejection rate' is an example of tangible standards. (True/False)

6.4 Critical Control Points

The managers have to exercise control by exception. They have to find out those deviations, which are critical and important for business. Minor deviations have to be ignored. Major deviations like replacement of machinery, appointment of workers, quality of raw material, rate of profits, etc. should be looked upon consciously. Therefore, it is said, "If a manager controls everything, he ends up controlling nothing." For example, if stationery charges increase by a minor 5 to 10%, it can be called as a minor deviation. On the other hand, if monthly production decreases continuously, it is called as a major deviation.

Broad classification of critical control points are as follows:

- **Physical standards** – Labour hours, speed of a machine, quantity of butter and sugar per kilogram of cake, etc. are physical standards. We have to use discretion on deciding which are critical. E.g., if sugar or butter is more or less, the cakes of Sambhavi will not be the same.
- **Cost standards** – You must be aware of the cost accountants and how they control costs. E.g., direct cost, indirect cost, labour cost per unit, material cost per unit - one kilogram cake, etc.
- **Capital standards** – This is important for a new project and also in the case of replacement of machinery. If the capital sunk is high, the fixed cost can be high and you may become very uncompetitive. For example, how much capital you can sink in a fruit shop depends on the place but it cannot be too high since you cannot increase the cost of fruits indefinitely. In case of Sambhavi Bakers, the capital standards for each outlet was put as Rs.200 lakh and for kid care facilities it was Rs.10 lakh.
- **Revenue standards** – This is a very common measure. Examples can be revenue per employee, revenue per customer, etc.
- **Programme standards** – If you want to introduce a new product or programme for improving the quality of the sales force, you can fix the standards not only in terms of cost but also success. E.g., to achieve 10% market share, ensure that 80% qualify in the sales force training, etc.
- **Intangible standards** – Intangible standards are not really immeasurable. They are of course difficult to measure. Standards such

as goodwill created by the salesperson, unit cohesion, learning of an individual, etc. are difficult to measure but there are methods to estimate most of these though they may not be as quantitatively measurable as profit or Return on Investment (ROI).

- **Goals as standards** – Organisations are using goals and objectives as standards. For example, in Management by Objective (MBO), the objectives are defined and some measure to evaluate it is created between the manager and the subordinate. The goal of a manager could be set as 'to open ten outlets in the next two years', 'reduce customer complaints by 5%,' etc. Here, the goals in conjunction with some measures are used for control.
- **Strategic plans and control points** – Can we control our strategy implementation? The answer is yes. If we have expansion strategy, we can define it as expanding business nationally or to three countries in Europe, etc. by a fixed date and these are evaluated.

Self Assessment Questions

5. The purpose of _____ is to focus the attention of the manager on things that are critical and important for business.
6. 'Labour hours' is an example of _____ standard.
7. Sambhavi Bakers put a standard that each outlet should not cost more than Rs.300 lakh. This is an example of setting _____ standard.
8. Management by objective is an example of using _____ as standards.

6.5 Control as a Feedback System

Management control is a feedback mechanism much like a thermostat in a refrigerator. What it does is to cut off the system when it reaches pre-defined level and then again switch it on when the temperature rises above a particular level. Figure 6.1 depicts the feedback loop of management control.

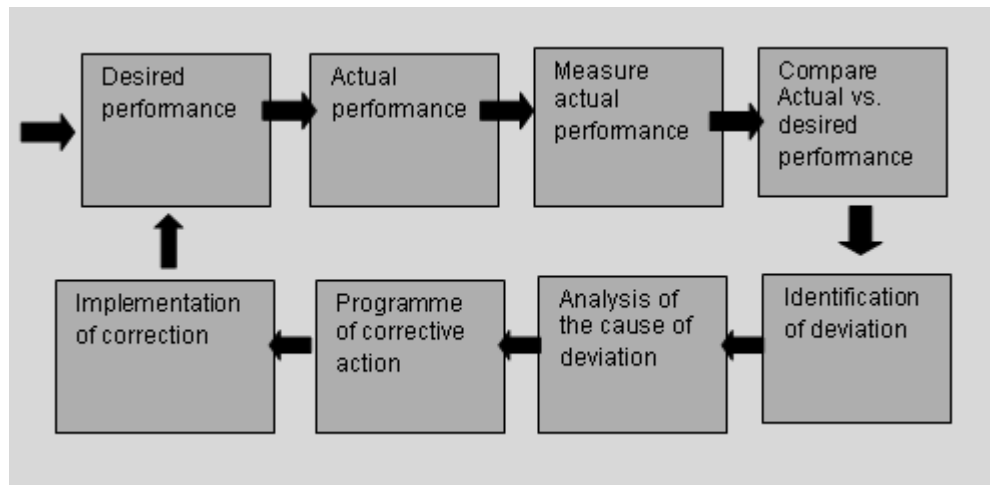


Fig. 6.1: Feedback Loop of Management Control

Real time information and control – Real time information about what is happening at a given time can be gathered in many cases. This is due to information technology. Some people see real time information as real time control. However, this is not so easy. Control requires analysis of the cause of deviation, which usually cannot take place in real time. Let us consider an example to understand this. If someone attaches you to an Electro-cardiogram (ECG) you will get real time information about the working of your heart; but if there is a deviation, it cannot be analysed and corrected real time. This analysis takes time. Similarly, Sambhavi Bakers may put in place a real time control system to check the delivery schedule of its products much like how a courier company has a tracking system on the net. But if the cake or letter is not delivered on time, the correction cannot be real time. It takes considerable time to analyse and correct. This does not mean real time information or prompt measurement of performance is not important.

From feedback to feed-forward – Take the case of driving a bike uphill. As you climb, you don't look at your speedometer and increase the speed when the speedometer reading falls. Instead, you keep accelerating after seeing the gradient. This is a classical case of feed-forward. In feed-forward, the system is focused on the input, which can create a variation in the output and can be corrected in time. It is not about post mortem but of proactive action. Most management systems fall short of it primarily because we tend

to use the financial data for management control i.e., the balance sheet data or the use of feedback from the input rejections in a factory to analyse, identify, and then correct the input or process. There are a very few mechanisms to do it proactively. However, we need to use feed-forward since it enables one to prevent mistakes.

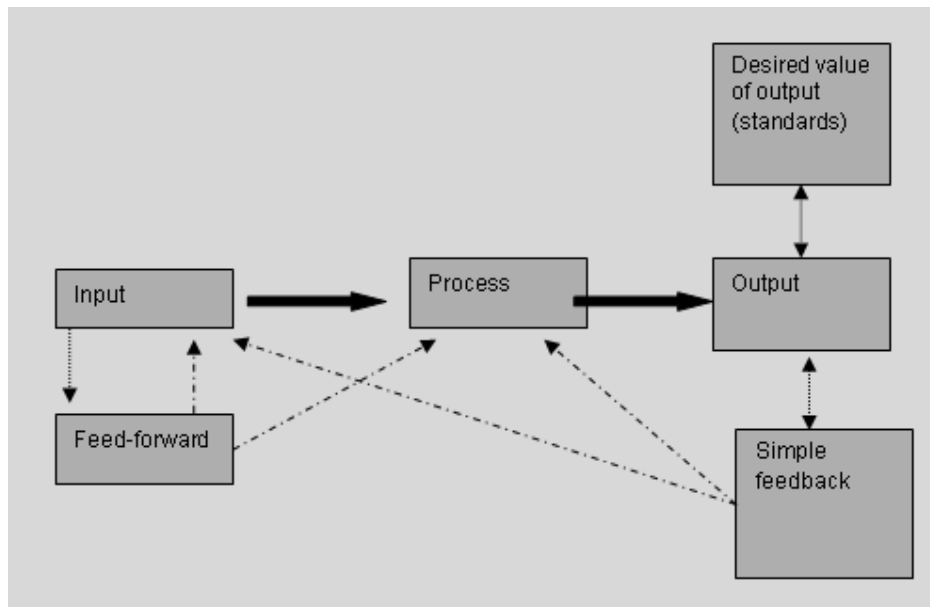


Fig. 6.2: Feedback – Feed-forward System- A Comparison

Requirements for feed-forward control – In order to create a feed-forward system, the following will be required:

- Identify important input variables.
- Develop a model for the system.
- Review regularly to see whether the input variables are identified and their inter-relationships continue to represent realities.
- Collect data on input variables regularly and put them into the system.
- Assess the actual input data regularly from planned-for inputs and evaluate the impact on the expected end results.
- Take action. While the system can indicate problems, the actions have to be taken by people. But these actions would prevent the problem rather than control the variations after it has happened.

Self Assessment Questions

9. Real time information always leads to real time control. (True/False)
10. Identifying important_____variable is the first step in creating feed-forward.
11. One of the reasons for management systems falling short of having good feed-forward is confining to the usage of _____data for creating the control systems.

6.6 Prerequisites of Effective Control

All managers like to have controls because without them their plans would go awry. Let us now study the pre-requisites to have an effective control system.

- ***Tailoring controls to plans and positions*** – A control is exercised on an activity or a group of activities. It follows that what control is good for a position may not be relevant for another e.g., the Vice President of marketing and the Vice President of operations cannot have the same controls though both maybe based on a financial control system. E.g., the outlet manager of Sambhavi can have a control based on ROI, but the snack counter manager needs to have it based on per customer revenue rather than ROI.
- ***Tailoring controls to individual manager*** – Controls have to be adjusted to the individual manager's capability also. If someone does not understand a control, he/she will not trust it or use it as a result of which it will become dysfunctional.
- ***Designing 'point to the exceptions at critical point'*** – If a control has to be effective, it must control the exception and that too at the critical point. For example, the critical point in home delivery of a birthday cake is the time and accuracy of writing the name. The exception can be wrong name due to spelling variations and the time of delivery due to wrong address. Therefore, control should exist so that the delivery order phone number (we usually experience it when we order a pizza) and the spelling of the name is rechecked. If the Assistant Manager can call the customer just before the delivery and recheck the delivery address and the spelling of the name, an appropriate control can be designed.

- **Objectivity of controls** – Many management actions are subjective, but when controls are created, they must be objective, accurate, and must suit a standard. While this may be relatively easy in machine related systems and financial related indicator, we have to be careful when we have to relate it to the intangible areas. For example, how would you have controls for the kid care manager in Sambhavi outlets?
- **Flexibility** – Controls must be flexible to include the changed plans, unforeseen circumstances, or outright failure. For example, Sambhavi may use budget control to say the inventory level but if the sales are significantly higher or lower, there should be flexibility in the control. Suppose we define the inventory per rupee of cost of production or per kilogram of production. It would be flexible to control it using an annual budgetary control or average inventory based on earlier data.
- **Fitting to the organisational culture** – Imagine putting tight control over Sambhavi whose culture is family-like and open with the freedom to experiment. The control will most certainly affect the culture which to begin with is the competitive advantage of Sambhavi. Therefore, it must fit the culture. If you have a tight and bureaucratic system, a loose control will also not work.
- **Economy of controls** – Controls must be worth their costs. Creating controls which are excessively expensive is counter-productive. For example, we cannot have the same controls in an aircraft and a car.
- **Ability to lead to corrective action** – The control should lead to corrective action. Only then it closes the loop and leads to better performance. For example, if the ROI of an outlet is below the standards specified, there should be a review system, which detects the sections that have not contributed their part and have exceeded it so that the performance can be corrected or rewarded.

Self Assessment Questions

12. Controls have to be common to all and should not be adjusted to the individual manager's capability (True/False).
13. A well designed control will point to the _____ at critical point.
14. Firm 'A' manufacturing gears follows an autocratic culture while Firm 'B' follows an open door policy and both have control measures. It will be a good idea that Firm 'A' borrows the 'control measures' from Firm 'B' to make the controls more effective. (True/False)

6.7 Control Techniques

Overall performance control

It is important to control the overall performance and not confine to some processes. Many overall controls in business are financial in nature. This is natural because money is the binding factor in business and can be easily used to get an integrated picture. Though it might have been more difficult in the days when computers did not exist, this is not so today. The financial controls used at the lower levels can be progressively aggregated upwards. Secondly, it also indicates what the goal achievement has cost in terms of resources. Thirdly, it is an excellent window to see the accomplishment of non-financial areas and the causes can be evaluated. E.g., the performance may be due to poor planning, inadequate training of employees, etc. The reasons for overall control are:

- Since overall planning is applied to the enterprise as a whole, control should also be applied to the entire organisation because planning without controlling is not an effective method as we saw in earlier sections.
- Decentralisation of authority is required especially in product and territorial divisions in semi independent units, which makes controlling important.
- Overall control permits the measurement of an integrated area manager's total effort rather than parts.

Overall performance control techniques

Let us now see some of the control measures.

- **Profit and loss control** – This is the simplest form and captures the revenue and cost. It can be made in perspective i.e., ahead of its happening by making a budget for the next year so that decisions can influence the revenues and expenses before they actually occur. This suffers from huge paper work involving transfers within the company but reduces when computers are used.
- **Control through Return on Investment (ROI)** – ROI measures both the absolute and the relative success of a company or unit by the ratio of earnings to investment on capital. This standard recognises that capital is the core of business.

- **Management audits and accounting** – Audit and accounting is another way to exercise overall control. Although they look at various financial measures, it is possible for an audit to evaluate the systems by asking penetrating questions on the financial indicators. Thus, for example, if the cost of procurement has increased substantially from the average in the last decade, then an audit process gives the feedback base on which the process itself can be controlled by taking corrective actions.
- **Bureaucratic and clan control** – Several organisations are controlled using elaborate rules and regulations, 'do it' instructions, etc. This is called bureaucratic control. Clan control is based on the norms, shared values, and expected behaviour. Most organisations have a combination of these to exercise control.

Balanced scorecard

One of the problems with the overall performance control is that the measures are all financial. Strategy based on which businesses are run is actually futuristic and therefore a financial measure based review is insufficient measure of performance. Therefore, we should measure other factors, which create the cohesion in the first place and then create financial results. These assets, though intangible, include customers and the internal process, which are the strengths of a firm, and learning and growth that essentially reside in the employees. If we do that, we would have taken care of internal factors. Customers and external factors are captured through financial data. Thus the four key components of a balanced scorecard are:

- Financial
- Customer
- Internal processes
- Learning and growth

This is created by Kaplan and Norton to measure the overall performance of a firm.

Budget as a control

A budget is a plan for a given period in numerical terms. They are statements of anticipated results. They may be in terms of financial figures, labour hours, materials, sales volumes, etc. If done with flexibility, they are excellent tools of control. Usually, budgeting is done by making incremental

changes to the existing budget, which is one of the reasons why it has got a bad name. Some of the popular budgeting methods are discussed below:

- **Capital budgeting** – Capital budget is the process in which a business determines whether projects such as building a new plant or investing in a long-term venture are worth pursuing. Often, a prospective project's lifetime cash inflow and outflow are assessed in order to determine whether the returns generated meet a sufficient target benchmark. Ideally, business should pursue all projects and opportunities that enhance shareholder value. However, since the capital availability at any given time for new projects is limited, management has to determine and decide which project yields the most return over a well defined period of time. This is done through capital budgeting. In other words, it controls the way the available resources are deployed.
- **Zero based budgeting** – It is usually done for support functions rather than production. In this method, every year, the activities and their costs are worked from the base. Hence every year the necessity of an activity has to be established. E.g., a training programme and its need have to be established every year and its cost also has to be estimated every year even if the programme itself is an old one. Thus, managers will think of new, more effective, perhaps less expensive faculty, venue, etc. This makes the manager think fresh every year about efficacy and cost.

Non-budgetary control

There are several non-budgetary controls. An inspection visit, managing by walking around, use of statistical data, benchmarking, operational audit, HR audit, etc. are the non-budgetary controls.

Benchmarking

Benchmarking is the process of comparing one's business processes and performance with industry bests and/or best practices from other firms or industries. Several measures are used for this. Quality, time and cost, employee per passenger, fuel consumed per passenger, etc. are benchmarks. In the process of benchmarking, management identifies the best firms in their industry, or in another industry where similar processes exist, and compare the results and processes of those studied (the "targets") with one's own results and processes. In this way, they learn how well the targets perform and, more importantly, the business processes that explain why these firms are successful.

Costing

- **Activity based costing** – When we allocate the cost of production to a product, we do it by the number of items made by a machine rather than the time taken for producing by the machine and its consequent cost. Sambhavi Bakers make cakes and biscuits in the same oven. Cakes are a low volume item and biscuits are made in thousands. However, cakes take a long time and are more complex to set up and operate than biscuits. Since the volume of biscuits is large, the oven makes biscuits most of the day. If we now allocate more variable cost to biscuits because it takes a long time, then it would result in an unfair advantage to the cost of the cakes. In other words, we will have no idea about the actual production cost of cakes. In order to avoid this, we use activity based costing in which the overheads and other costs attributable to the activity of baking cakes is allocated to cakes appropriately.
- **Target costing** – One way to price a product is to make it, identify the cost incurred, and then fix its actual price. But that price may not be competitive. Target costing involves setting a target cost by subtracting a desired profit margin from a competitive market price. In a more formal way we call target costing as "Target Costing is a disciplined process for determining and achieving a full-stream cost at which a proposed product with specified functionality, performance, and quality must be produced in order to generate the desired profitability at the product's anticipated selling price over a specified period of time in the future." These concepts are supported by the four basic steps of target costing:
 - 1) Define the product
 - 2) Set the price and cost targets
 - 3) Achieve the targets
 - 4) Maintain competitive costs

Target costing controls the cost by forcing the managers to innovate design, input, processes, etc. so that the target is achieved.

Through quality tools/logistic control

- **Total Quality Management (TQM)** –
 - 1) It is the belief that quality can and must be managed.
 - 2) The first realisation takes you the next step i.e., the problems are created by process and not by people. So this control makes you look at the processes rather than employees.

- 3) One should look for a cure rather than treat the symptoms and thus control the quality over a long period of time.
- 4) Every employee is responsible for quality and therefore makes quality control more pervasive or spreads throughout the company.

It then states that quality control should be measurable and therefore creates a measure and then goes on to make the improvement continuous. It makes quality a long term investment and thus brings an effective control on the product lines, processes, and also overall performance.

- **Kaizen** – Kaizen is a philosophy based on team work, personal discipline, improved morale, quality circle, and suggestion for improvement. It has three foundations namely elimination of waste and inefficiency (also called muda in Japanese), Kaizen 5 S framework for good housekeeping and standardisation. The 5 S framework means tidiness (seiri), orderliness (seiton), cleanliness (seiso), standardised clean up (seiketsu), and discipline (shitsuke). As one can see, all of them contribute to control the quality, waste, cost, etc. and hence to become an effective control tool.
- **Just in Time (JIT)** – Any inventory whether of parts or finished goods is a cost and must be controlled. JIT is a plan to ensure that the inventory is reduced to almost nil and it is a plan that Toyota tried out successfully. The concept of control by JIT is not very different from controlling the inventory. JIT differs from inventory control in its philosophy to keep the required parts manufacturer and the outlet which sells well integrated into the main plant to include the quality of the supplies. Thus it is possible to get the inventory right into the shop floor without going to the store on a daily/hourly basis. This obviously controls cost. Needless to say, that it needs good discipline at all levels to apply this. JIT controls by eliminating the following:
 1. Waste from overproduction
 2. Waste of waiting time
 3. Transportation waste
 4. Processing waste
 5. Inventory waste
 6. Waste of motion
 7. Waste from product defects

- **Six sigma** – Six sigma stands for six standard deviations (Sigma is the Greek letter used to represent standard deviation in statistics) from the mean. Six sigma methodologies provide the techniques and tools to improve the capability and reduce the defects in any process. It was started in Motorola, in its manufacturing division, where millions of parts were made using the same process repeatedly. Six sigma methodologies improve any existing business process by constantly reviewing and re-tuning the process. To achieve this, Six sigma uses a methodology known as DMAIC (Define opportunities, measure performance, analyse opportunity, improve performance, control performance). Six sigma strives for perfection and allows for only three to four defects per million opportunities for each product or service transaction. It relies heavily on statistical techniques to reduce defects and measure quality.

Through time event network analysis

There are many such methods. Let us have a look at a few of them.

- **Gantt charts** – A Gantt chart is a type of bar chart, developed by Henry Gantt, which illustrates a project schedule. Gantt charts illustrate the start and finish dates of elements of a project. The terminal elements control process. If a process requires three months and we know when it should be completed, then we can plot when it should start and when each of the sub-activities should finish. Figure 6.3 depicts a simple Gantt chart.

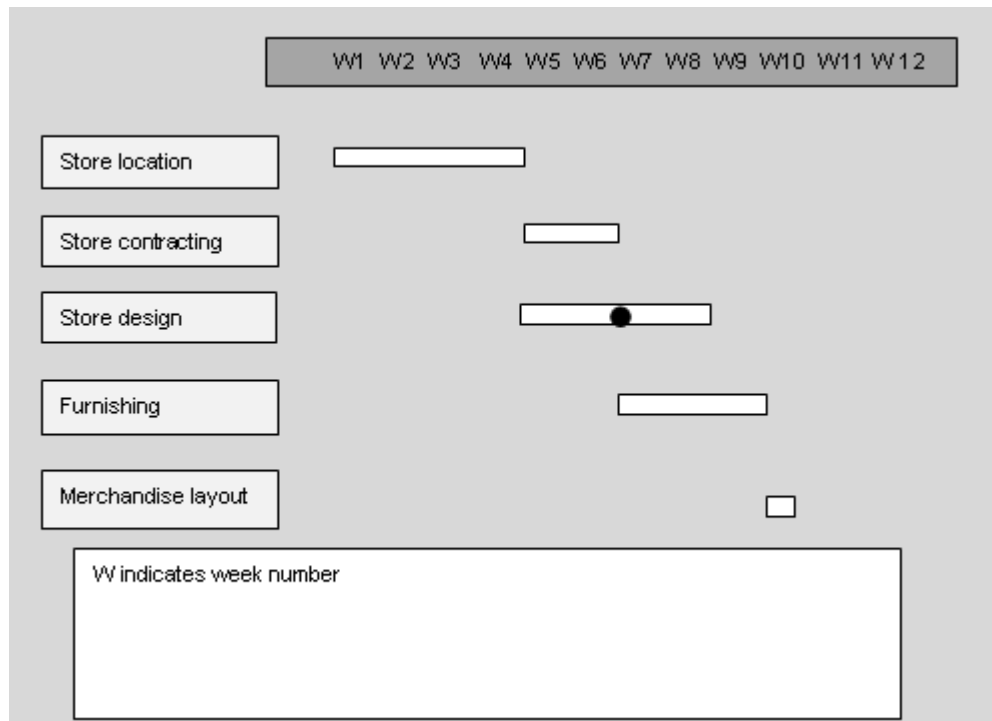


Fig. 6.3: Gantt Chart

This shows how a 12 week store opening of Sambhavi Bakers can be planned using a Gantt chart. It can be seen that locating the store ends by 4th week and then contracting starts. While it is going on, the design can be started and while design is half way the furnishing can be started (rather than wait for the design to be completed) and only after furnishing we can start to merchandise the layout.

Gantt charts can have more details, e.g., milestones i.e., the milestones can be fixed along with the Gantt chart. For example, the black bullet point indicates a milestone since furnishing can start only after crossing this. If there is a delay in reaching the milestone, subsequent activities will also be delayed and hence the importance to monitor design.

- **Program Evaluation and Review Technique (PERT) and Critical Path Method (CPM)** - A PERT chart is a graphic representation of a project's schedule. It shows the sequence of tasks, which task can be performed simultaneously. It also shows the critical path of tasks that must be completed on time so that the project meets its completion

deadline. The chart can be constructed with a variety of attributes, such as earliest and latest start dates for each task, earliest and latest finish dates for each task, and slack time between tasks. A PERT chart can document an entire project or a key phase of a project. The chart allows a team to avoid unrealistic timetables and schedule expectations, to help identify and shorten tasks that are bottlenecks, and to focus attention on most critical tasks.

Steps in creating a PERT chart are as follows:

- Identify the specific activities and milestones
- Determine the proper sequence of activities
- Construct a network diagram
- Estimate the time required for each activity
- Determine the critical path
- Update the project progresses

Advantages of PERT are:

- It forces the managers to plan since they have to make a time event chart.
- Forces planning all the way down the line because each subordinate manager must plan the event for which he/she is responsible.
- Concentrates attention on critical elements that may need correction.
- Makes forward-looking control possible.
- Enables managers to aim reports and builds pressure for action at the right spot.

The disadvantages are:

- When a programme is new or ambiguous, and no reasonable estimate of time can be made, PERT is difficult to implement.
- It emphasises only on time and not cost.

Let us now create a PERT and CPM model for opening a store of Sambhavi Bakers as given in Figure 6.4. The activities and their numbers are given below. The corresponding activity number is given in a shaded circle in the PERT chart. The arrows indicate the path of each activity and which activity has to follow which. The critical path is traced through the thick arrow.

1. Identify the general area for the stores

2. Identify the possible stores
3. Complete negotiation with owners
4. Complete legal contacting
5. Commence designing of the stores
6. Confirm design based on the final selection of stores
7. Order furnishing
8. Complete furnishing
9. Order merchandise
10. Place merchandise
11. Advertise
12. Open stores

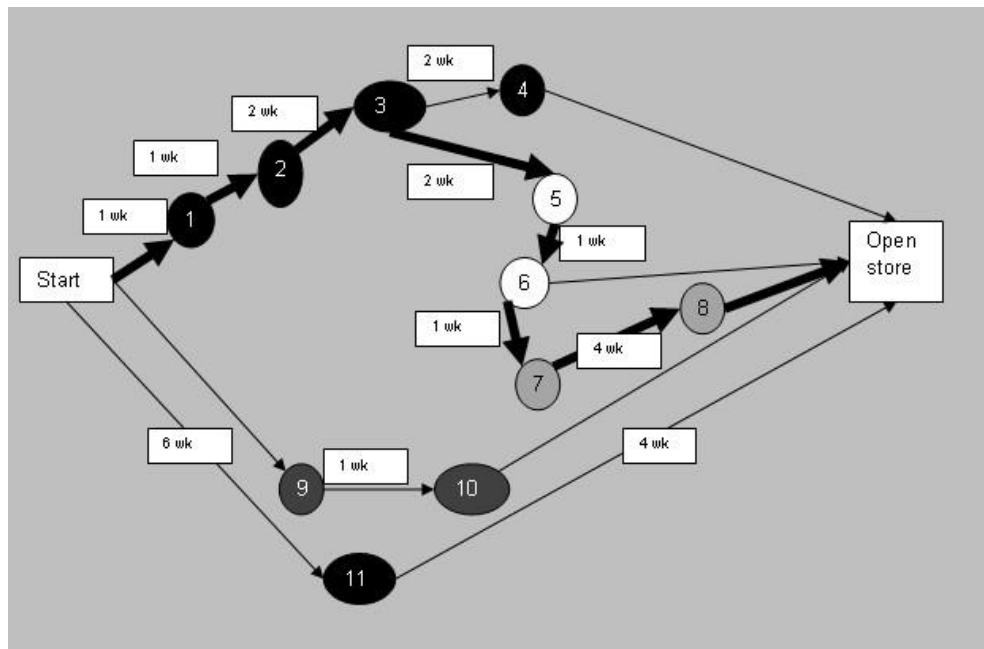


Fig. 6.4: PERT and CPM Model of Sambhavi Bakers

Thick black arrows in the figure 6.4 shows the critical path i.e., if anything goes wrong on this path, the opening will be delayed. The activities along the thin arrow can be done well within the time the activities along the thick arrows are done. Hence, these are not the critical paths. Time for each activity is given as weeks (1 week, 6 week, etc) in boxes.

Self Assessment Questions

15. Many overall controls in business are financial in nature. This is natural because money is the _____ in business and can be easily used to get an _____ picture.
16. Clan control means using controls based on the _____, _____ and expected behaviour.
17. Balanced scorecard is an overall performance control method and measures financial, customer, internal processes, and _____ factors.
18. Usually, resources available to an organisation at any time are scarce and opportunities are plenty. Investing in projects that gives the best return over a long period can be done by----- --.
19. Zero based budgeting enables managers to create a budget quickly by adding or subtracting to earlier expense on this account. (True/False)
20. Benchmarking must ensure that the best practices taken into consideration for benchmarking are from the same industry and not from another industry. (True/False)

6.8 IT Enabled 'Controls' and its Challenges

In the earlier section we learnt about real time controls. Most real time controls are microprocessor or IT enabled. But unlike earlier times, IT enabled controls create their own unique challenges to business.

Some of the far reaching IT enabled processes, which have facilitated control are as follows:

- **Enterprise resource planning system** – It is a modern system of integrating the manufacturing or service process upward and downward. It connects the logistics, sales, HR, finance, and other functions in an integrated and real time manner. So, though we may still need time to analyse the variance, the prompt report system enables us to reduce the reaction time and thus enables better control.
- **Speech recognition software** – This software is still far from being a fully developed product. When done, it would perhaps be possible to pass instructions to a computer and then on to a machine much like you pass instructions to a worker. Combined with mobile communication systems, this would be able to exercise real time control on a remote

basis, while on the move and perhaps even when we are doing something else. Voice control will also make control more pervasive. Care must be taken so that it does not interfere with empowerment.

- **Telecommuting** – Using the company's main server and communication, employees are able to work from home or any other place as effectively as one can do from office. This would once again enhance the capability to exercise control because you can remain connected to the workplace almost from anywhere.
- **Computer networks** – Networked computers make it easy for everyone to see and gain the input at the same time. It also enables one to see the contents of another computer and exercise control of the workplace. E.g., a person who is doing medical transcription can be controlled from a remote location by a supervisor by merely looking into his/her system.
- **Internet, intranet, and extranet** - Though the Internet is the most pervasive, the other networks are also important components in creating controls. For example, in marketing, the control that the sales person or the distributor had over the customer is now gone because it is possible for a person to consult a social network and purchase an item without any of the traditional channels interfering. On the other hand, the control that the business has over the market has also increased by making a product available on a universal basis. For example, if you have a system of ordering cakes of Sambhavi bakers online and a system of delivering, the control exists for both the parties.
- **Groupware** – This enables a document to be seen by several people simultaneously. This implies that it is possible to design something sitting in multiple locations much like how Boeing designed one of its planes from 245 different locations. This means that close control is possible from multiple locations unlike earlier days.
- **E-commerce and M-commerce** – There are several control measures that take the form of E-commerce and M-commerce. Transfer of money is one thing, verification of details before approval such as electronic verification and signature is another. Use of Radio Frequency Identity (RFID) (identifying a cargo by embedding a chip that transmits a particular number enables you to control the move of the cargo), supply of materials and goods on JIT basis, reducing inventory, control of

tenders, etc. are other examples. It is a subject itself that the intention here is only to highlight the existence and potential of the system.

Activity 1:

Study the PERT and CPM diagram given in the text and identify what is the minimum time that would be required to complete opening a new outlet by Sambhavi Bakers.

Activity 2:

Read the following situation and prepare a report.

On 1 January, 2012, a person who is running a bakery in a large town called up the Project Manager of Sambhavi Bakers. He told the manager that he wishes to sell his store as he is old and is now having advanced stage of cancer and that his children do not want to continue. He had served the area well and everyone liked his bakery and he wanted the place to continue as a bakery. The area as per the description is large and has adequate space for the Sambhavi model. The Chairman gave the go-ahead on 1 February, 2012 to start operation in the new town. Assuming that Sambhavi has a reputation for fast implementation of projects, when do you think that the outlet can be opened?

Self Assessment Questions

21. Telecommuting enhances the capability to exercise control because you can remain connected to the _____ almost from anywhere.
22. Enterprise Resource Planning (ERP) system is effective not only because it is a prompt report system and reduces reaction time but also because there is no need to analyse the cause of the variance. (True/False)

6.9 Summary

Let us recapitulate the important concepts discussed in this unit:

- Controlling can be defined as measuring and correcting of performance to achieve the organisational goals.
- The process of planning and controlling works on systems approach, Planning → Results → Corrective Action (through control)

- Controlling is important as it creates basis for the future, ensures goals are on track, prevents repetition of mistakes, enables future planning, enables decentralisation, assist co-ordination, enables motivation and management by exception.
- Steps involved in controlling are establishment of standards, measurement of performance, comparison of actual performance and standard performance, taking remedial actions.
- Critical control points may be classified as physical standards, cost standards, capital standards, revenue standards, programme standards, intangible standards, goals and strategic plans.
- Feedback loop of management control consists of desired performance, actual performance, measuring actual performance, comparing with desired performance, indentifying deviations, analysing causes of deviation, planning corrective action and implementing correction.
- Effective control must be tailored to plans and positions, individual manager, objective, flexible, fitting to the organisational culture, economical and must lead to corrective action.
- Some of the overall control techniques are profit and loss control, control though ROI, audits and accounting, bureaucratic and clan control.
- A balanced scorecard was created by Kaplan and Nortan to measure the overall performance of a firm.
- Some of the quality tools of control are TQM, Kaizen, JIT, Six Sigma, Gantt Charts, PERT and CPM.

6.10 Glossary

Balanced scorecard: A technique of overall performance control created by Kaplan and Norton to measure and control strategy and its implementation. It measures financial, customer, internal process, and learning and growth of an organisation.

Benchmarking: A process of comparing own standards and best practices in an activity or process with that of another in the same or different industry. For example, to benchmark practices of recovery of debts, you can benchmark with another firm in your own industry or another industry.

Critical control points: Selection of important points in a process or activity for measuring because of the impact on the overall process. By controlling this point, most of the important variations can be controlled.

Critical path: A term used in PERT and CPM methods of project management. It is that series of activities in sequence (path) along which if there is any delay the project will get delayed. The other paths are usually shorter or flexible enough to be absorbed within the time taken by the critical path of activities.

Overall performance control: Organisation has several activities, units, etc. Many controls pertain to only the activity or unit. Those measures and standards that control the overall performance of an organisation rather than its individual units, departments, or activities are called overall performance control.

6.10 Terminal Questions

1. Explain the controlling process.
2. Why is controlling important in management functions?
3. What are the different critical control points?
4. What are the prerequisites of effective control?

6.11 Answers

Self Assessment Questions

1. measuring and correcting
2. Goals
3. False
4. True
5. Critical control point
6. Physical
7. capital
8. Goals
9. False

10. Input
11. financial
12. False
13. Exceptions
14. False.
15. Binding factor, integrated
16. Norms, shared values
17. Learning and growth
18. Capital budgeting
19. False
20. False
21. Workplace
22. False

6.12 Terminal Questions

1. Process of controlling includes establishment of standards, measurement of performance, comparison of actual and standard performance and taking remedial action. Refer Section 6.3 for more details.
2. Controlling is important because it keeps goals on track and creates the basis for the future. Refer Section 6.3 for more details.
3. The different control points are physical standards, cost and capital standards, revenue standards, programme standards, intangible standards etc. Refer Section 6.4 for more details.
4. Some of the prerequisites of effective control are flexibility, fitting to the organisation culture, tailoring controls to plans, positions and to the individual managers. Refer Section 6.6 for more details.

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